F. Digit Lock

time limit per test: 1 second

memory limit per test: 256 megabytes

Alice has a treasure secured by a 4-digit numeric lock.

The 4-digit numeric lock.

Of course, Alice knows the correct combination C required to unlock this lock. However, there is an issue. Due to a mechanical fault in the numeric lock, certain combinations are deemed **forbidden**.

Alice wants to unlock this lock. In a single step, Alice can take one of the four "wheels" and rotate it by one step. More formally, Alice can take a single digit and either increment it or decrement it by 1 (modulo 10). The digits rotate around like clocks, so if we decrement 0 the digit becomes 9 and if we increment 9 the digit becomes 0. See the sample explanation for better understanding.

Alice wants to find the minimum number of steps required to reach the correct combination C from the starting combination S.

Input

The first line of the input contain the starting combination S and the correct combination C. Both are described in the input as 4-digit numbers. It is guaranteed that the starting combination is not forbidden.

The second line contains an integer n ($1 \leq n \leq 10^4$), the number of forbidden combinations.

The next n lines each contain a forbidden combination.

Output

If it is impossible to reach C, then output -1. Otherwise, output the minimum number of required steps.

Examples

input Copy

```
0000 0003
```

```
1
0001
```

output

```
5
```

input

```
0000 0007
1
0006
```

output

```
3
```

input

```
1234 5678
12
2234
3234
4234
5234
1334
1434
1534
1634
1244
1254
1264
1274
```

output

```
16
```

Note

In the first sample, our starting combination is 0000 and our target combination is 0003, while there is only one forbidden combination: 0001. So we can not directly increment the last digit in our starting combination. The following steps are optimal:
$0000 \to 0010 \to 0011 \to 0012 \to 0013 \to 0003$.

in the second sample, our starting combination is again 0000 while our target combination is 0007. The following steps are optimal: $0000 \to 0009 \to 0008 \to 0007$.